



TECHNICAL DESIGN SUBMISSION

# अनुवादिनी

Anuvadini — Offline Translation Suite

EVALUATION JURY

HSBC Technical Review Committee

TARGET DEPLOYMENTS

BAIF Development Research Foundation

SYSTEM ARCHITECTURE

Offline-First CPU Optimization



# | Problem Statement & Field Context



## Language Barriers in Rural Extension

Field trainers require rapid localization of audio/video manuals from English & Hindi into regional languages (e.g., **Marathi**) for offline deployment.

🚫 **No-GPU Dependency:** Must run reliably on existing low-spec single-core CPU desktops in Pune/BAIF offices.

**Disconnected Field Reality:** Internet is only for installation; translations happen strictly offline before traveling to remote regions.

🤖 **Strict Zero-Cost Mandate:** Absolutely zero software/licensing budgets. Must utilize open-source frameworks exclusively.



# अनुवादिनी Core Pipeline Design

The application orchestrates a continuous, localized translation loop running entirely on CPU. It parses inputs, processes them across isolated model layers, and yields synchronized media deliverables.

**Step 1**

## Media Intake

Accepts MP4, MKV, MP3, WAV, etc. segments up to 15 mins. Processes audio extraction using FFmpeg subprocesses.

**Step 2**

## STT Engine

Launches Faster-Whisper (INT8 quantized). Decodes speech with localized dialect filters on single-thread CPU cycles.

**Step 3**

## NMT Translation

NLLB-200 engine maps Marathi and Hindi semantics. Quantization offsets memory use below 2GB RAM.

**Step 4**

## Asset Delivery

Compiles localized audio files via eSpeak NG, synthesizes SRT/ASS captions, and burns subtitles using local FFmpeg layers.



# Proposed Open-Source Tech Stack

Custom quantization transforms demanding LLM/NMT models into lightweight structures executable on base corporate PCs.

| Task Category         | Open-Source Standard              | Offline Optimization Strategy                                                                                       |
|-----------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| ASR (Speech-to-Text)  | Faster-Whisper (Medium-Quantized) | Intel MKL & CTranslate2 orchestration reduces raw RAM footprint down to ~1.2 GB, boosting execution times.          |
| NMT (Translation)     | NLLB-200 (No-Cloud Model)         | Optimized parameters distilled weights specifically for regional Marathi syntax and grammatical patterns.           |
| TTS (Audio Synthesis) | eSpeak NG Synthesis Engine        | Non-autoregressive model that converts text back into natural regional voiceover output with instant CPU rendering. |
| Media Multiplexing    | FFmpeg Pipeline                   | Compiles and synchronizes translated voice track overlays and burns ASS captions directly onto output MP4s.         |



# | Compliance with Evaluation Pillars (1)



## 1. Impact & Reach

Ensures vital farming guidance (soil health, animal husbandry) reaches Marathi and Hindi-speaking communities, democratizing agricultural extension.



## 2. Scalability

Slices dependencies down to single CPU setups. Requires no active GPU clusters, making the software distributable via USB drives to any regional sub-office.



## 3. Production Readiness

Built-in dynamic audio segmentation allows the app to process files up to 15 minutes without running out of standard RAM or system cache.



## | Compliance with Evaluation Pillars (2)



### 4. Usability / UX

Simplified file portal mimics the provided mockups: clear drag-and-drop mechanics, side-by-side editing review panels, and single-click transcript downloads.



### 5. Tech Safety

100% on-premises execution guarantees zero data leaks or external network breaches. Preserves native scripts for Marathi and Hindi translation accuracy.



### 6. Zero Cost (OS)

Relies strictly on open-source weights. No paid OpenAI, Microsoft Azure, or AWS cloud APIs. Free of ongoing software licensing overhead.

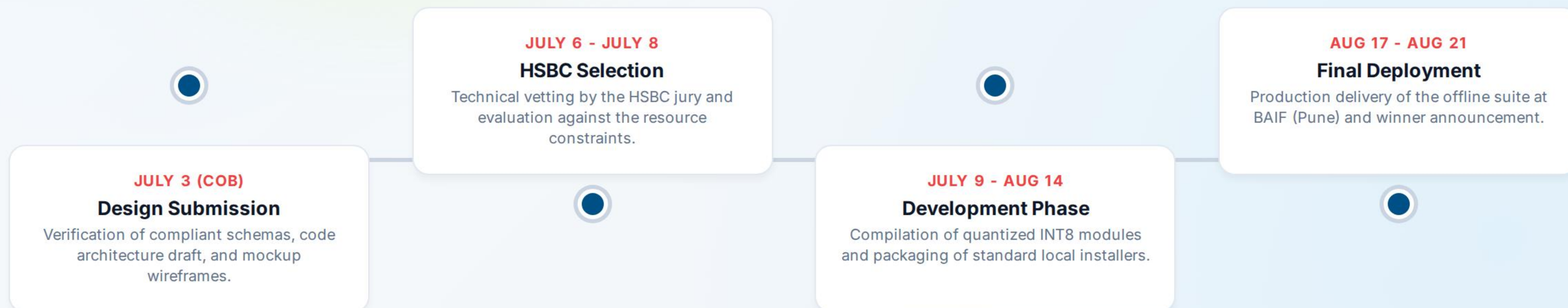


### 7. Optimization

Leverages CTranslate2 compilation pipelines and INT8 quantization matrices, ensuring the heaviest translation layers render in real-time.



# | Project Implementation Roadmap





# | Image Sources



[https://images.splitshire.com/full/Ramu-the-Farmer:-A-Kid-Friendly-Animated-Village-Morning\\_Prre3.png](https://images.splitshire.com/full/Ramu-the-Farmer:-A-Kid-Friendly-Animated-Village-Morning_Prre3.png)

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